

CPEN 416 / EECE 5710

Lecture 19: Intro to quantum error correction

Tuesday 18 November 2025

Announcements

- Quiz 8 today
- Final week schedule updates
 - Mon. 1 Dec: last tutorial, exam review
 - Tues. 2 Dec: quiz 10, groups 01-02 presentations
 - Thurs. 4 Dec: groups 03-05 presentations

Last time

We defined *quantum channels*,

Quantum channels are CPTP linear maps.

- **T**race-**P**reserving:
- Positive:
- **C**ompletely **P**ositive:

Last time

Quantum channels are characterized by **Kraus operators** $\{K_i\}$,

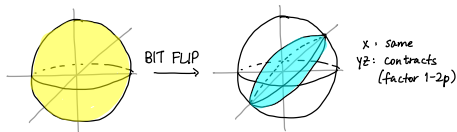
Example: unitary channel, $\mathcal{U}$, $\{K_i\} = \{U\}$.

Example: a projective measurement $\mathcal{M}$, $\{K_i\} = \{\Pi_i\}$

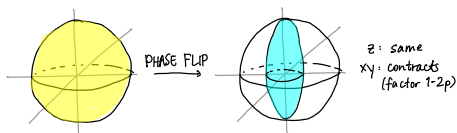
Last time

We expressed a noisy processes as quantum channels.

Bit flip channel

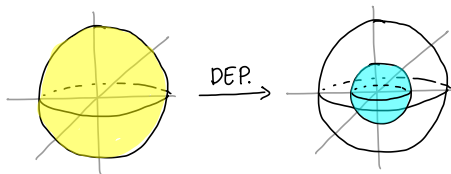


Phase flip channel



Last time

Depolarizing channel



Amplitude damping channel

Last time

We compared density matrices using two distance measures.

Trace distance

Fidelity

We simulated these processes and computed distance measures using PennyLane's ‘`default.mixed`’ device.

Learning outcomes

Module 5: introduction to quantum error correction

Today: the basics by example

- 1 Encode quantum information into a logical qubit using a repetition code
- 2 Design circuitry to correct bit flips and phase flips using repetition codes
- 3 Correct bit and phase flip errors with the 9-qubit code

Next time: formalizing today's ideas

Quantum error correction

Our current picture of noise:

We can protect qubits against noise using error correcting codes.

Repetition codes

Imagine a bit string transmitted through a classical channel that flips each bit (individually) with probability p .

Idea: add redundant information to enable *detection* and *correction* of bit flips.

Define *encoding* and *decoding* operations, $\mathcal{E}$ and $\mathcal{D}$,

Repetition codes

Devise a procedure, $\mathcal{R}$, to recover from an error: majority voting.

Operation	Outcome
	0 0 1 0 1
$\mathcal{E}$	
ϕ	
$\mathcal{R}$	
$\mathcal{D}$	

Is this better? When will this fail?

Quantum repetition code (bit flip code)

Idea:

Why won't this work?



Alternative:

Bit flip code: encoding circuit

Exercise: (a) How does $\mathcal{E}$ affect a general state,

$$\mathcal{E}(\alpha|0\rangle + \beta|1\rangle)$$

(b) What does the corresponding circuit look like?

Designing logical operations

Logical operations are specific to codes.

They should act on the logical states $|0\rangle_L$, $|1\rangle_L$ the same way the physical operations act.

Exercise: design circuits for logical X , Z , H , S , and $CNOT$.

Bit flip code: recovery operation

Let's design a circuit to detect (a) *whether* an error occurs, and
(b) *which* error occurs.

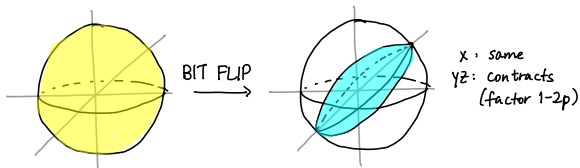
Bit flip code: recovery operation

Design a circuit to *recover* the state of the first qubit

Bit flip code: full circuit for bit flip code

Bit flip code: analysis

Imagine any possible state that can go through the channel:



Exercise: Determine $F(|\psi\rangle, \Phi(|\psi\rangle \langle\psi|))$ where Φ is the *bit flip channel* with parameter p .

Bit flip code: analysis

Compare overall fidelity of the state *after the error*, vs. fidelity *after recovery*.

After the error

After recovery, have the state

Phase flip errors

With our encoding

and appropriate circuitry, we can correct *bit flip errors* but not phase flip errors.

Phase flip code: encoding circuit

Main idea: make phase flip errors “look like” bit flip errors.

Shor code

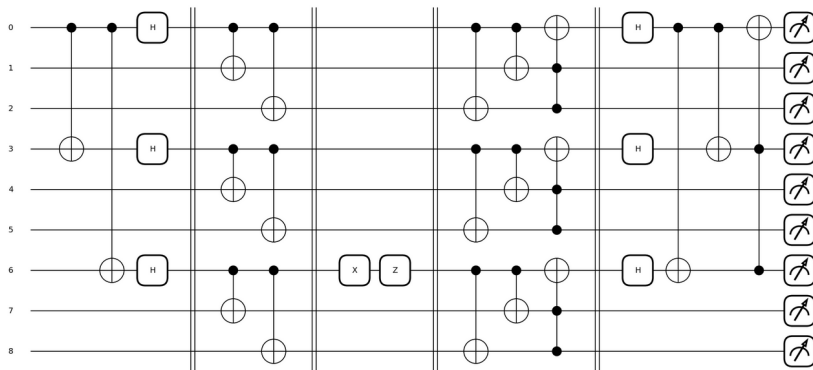
To correct a combination of one bit flip and/or phase flip error, we can *concatenate* codes.

Shor code

The Shor code can correct one *arbitrary* error on a *single* qubit.

Imagine a bit + phase flip on qubit 6:

Shor code



Next time

Next class:

- Properties of errors and error correcting codes
- Stabilizers and stabilizer codes

Action items:

- ① Work on project

Recommended reading:

- From this class: Codebook EC; N&C 10.1-10.2
- For next class: Codebook EC; 10.3, 10.5